

Number theory – 8

Pell equations, Markov-type equations and Vieta jumping

24 сентября 2026 года

Theoretical minimum. A quadratic equation in two variables is solved in the ring $\mathbb{Z}[\sqrt{d}]$: the solutions are finitely many elements times the powers of one unit, read off from a continued fraction. An equation quadratic in each variable separately is solved by Vieta jumping: replace a root by the other one and descend to finitely many fundamental solutions. Existence of solutions of $x^2 - dy^2 = 1$ is Number theory – 5.

Definitions. For irrational α put $\alpha_0 = \alpha$, $a_k = \lfloor \alpha_k \rfloor$, $\alpha_{k+1} = 1/(\alpha_k - a_k)$; then $\alpha = [a_0; a_1, a_2, \dots]$, and the convergents $p_k/q_k = [a_0; a_1, \dots, a_k]$ satisfy $p_k = a_k p_{k-1} + p_{k-2}$, $q_k = a_k q_{k-1} + q_{k-2}$, $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}$. For a non-square $d > 1$ the norm $N(x + y\sqrt{d}) = x^2 - dy^2$ is multiplicative on $\mathbb{Z}[\sqrt{d}]$; units are the elements of norm ± 1 , η is the least unit > 1 and ε the least solution > 1 of $x^2 - dy^2 = 1$, so $\varepsilon = \eta$ or η^2 . A Markov triple is a solution of $x^2 + y^2 + z^2 = 3xyz$ in positive integers. A Vieta jump replaces a root z of $z^2 - sz + t = 0$ by the other root $z' = s - z = t/z$.

Theorem 1 (Legendre). $|\alpha - \frac{p_k}{q_k}| < \frac{1}{q_k q_{k+1}}$, and conversely $|\alpha - \frac{p}{q}| < \frac{1}{2q^2}$ implies that p/q is a convergent. Hence if $0 < |x^2 - dy^2| < \sqrt{d}$ with $x, y > 0$, then x/y is a convergent of $\sqrt{d}$: for $n = x^2 - dy^2 > 0$ one has $0 < \frac{x}{y} - \sqrt{d} = \frac{n}{y(x+y\sqrt{d})} < \frac{1}{2y^2}$, and for $n < 0$ the same works for y/x and $1/\sqrt{d}$.

Theorem 2 (periodicity and units). A continued fraction is eventually periodic exactly for quadratic irrationals (Euler, Lagrange), and $\sqrt{d} = [a_0; \overline{a_1, \dots, a_{r-1}, 2a_0}]$. With $\alpha_k = (P_k + \sqrt{d})/Q_k$ one has $p_{k-1}^2 - dq_{k-1}^2 = (-1)^k Q_k$, $0 < Q_k < 2\sqrt{d}$, and $Q_k = 1$ iff $r \mid k$. The units are $\pm \eta^k$, $k \in \mathbb{Z}$. *Proof:* a unit $u = x + y\sqrt{d} > 1$ has $|\bar{u}| = 1/u < 1$, so $x, y \geq 1$; if $\eta^k \leq u < \eta^{k+1}$, the unit $u\eta^{-k} \in [1, \eta)$ equals 1.

Theorem 3 ($x^2 - dy^2 = n$). For $n \neq 0$ call solutions equivalent if one is $\pm \varepsilon^k$ times the other. Each class contains $\beta = u + v\sqrt{d}$ with $\sqrt{|n|} \leq \beta < \varepsilon\sqrt{|n|}$, so there are finitely many classes, say h , and $\frac{h}{\ln \varepsilon} \ln X + O(1)$ positive solutions with $x \leq X$. Congruences (Number theory – 3) often prove $h = 0$, but not always (Problem 1).

Theorem 4 (the Markov tree). If $x \leq y \leq z$ is a Markov triple, then $z' = 3xy - z = (x^2 + y^2)/z$ is a positive integer, and $z' < z$ unless $x = y = z = 1$: the quadratic $t^2 - 3xyt + x^2 + y^2$ is negative at $t = y$ once $(x, y) \neq (1, 1)$. So all Markov triples descend to $(1, 1, 1)$ and form a binary tree, (x, y, z) with $x < y < z$ having the children $(x, z, 3xz - y)$ and $(y, z, 3yz - x)$; Markov numbers are $1, 2, 5, 13, 29, 34, 89, 169, 194, \dots$. Likewise for the Hurwitz equation $x_1^2 + \dots + x_n^2 = ax_1 \dots x_n$ a solution with $x_n > \frac{a}{2}x_1 \dots x_{n-1}$ jumps down, so all solutions come from finitely many fundamental ones (descent: Methods of proof – 2).

Fact 1 (counting). The number of Markov triples with $z \leq N$ is asymptotic to $C(\ln N)^2$, $C \approx 0.18$ (Zagier), against $c \ln N$ solutions of a Pell equation. Whether z determines the triple is Frobenius' uniqueness conjecture, still open.

1. Prove Theorem 1, including Legendre's criterion. Using $\sqrt{37} = [6; \overline{12}]$, find all integers n with $0 < |n| < \sqrt{37}$ for which $x^2 - 37y^2 = n$ is solvable, and show that $x^2 - 37y^2 = 3$ has no integer solution although it is solvable modulo every positive integer m .
2. Let a and b be positive integers such that $ab + 1$ divides $a^2 + b^2$. Prove that $\frac{a^2 + b^2}{ab + 1}$ is the square of an integer. (IMO 1988)
3. Let a, b, c be positive integers with $a^2 + b^2 + 1 = abc$. Prove that $c = 3$, and find all such pairs (a, b) . (IMS 2014)
4. Prove that if p and q are rational numbers and $r = p + q\sqrt{7}$, then there exists a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ with integer entries and with $ad - bc = 1$ such that $\frac{ar + b}{cr + d} = r$. (IMC 2005)
5. Let a be a positive integer and $n \geq 4$. Prove that $x_1^2 + \dots + x_n^2 - ax_1x_2 \dots x_n \leq 0$ for all real numbers with $2 \leq x_1 \leq \dots \leq x_n \leq \frac{a}{2}x_1 \dots x_{n-1}$, and find the cases of equality. What does this say about the fundamental solutions of the Hurwitz equation $x_1^2 + \dots + x_n^2 = ax_1 \dots x_n$? (IMC 2025, shortlist)